MindTrace AI: Exploring the Trade-Off Between Efficiency and Cognitive Depth in AI-Assisted Learning

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ABSTRACT

This study examines how AI-assisted systems influence human learning behaviour, with a focus on the trade-off between performance efficiency and conceptual depth.

A quasi-experimental design ($n = 18$, ages 14–18) compared AI-assisted learners ($n = 10$) with independent learners ($n = 8$) across task completion time, accuracy, conceptual depth, and engagement.

Results indicate that AI assistance significantly improves efficiency (7.2 vs 13.6 minutes, $p < 0.01$) and accuracy (91% vs 68%), while independent learning demonstrates higher conceptual depth (7.8 vs 6.5).

These findings suggest that AI systems optimize short-term performance by reducing cognitive load, but may limit deeper cognitive processing when over-relied upon.

The study proposes a hybrid interaction model, where AI is used as a feedback mechanism rather than a primary reasoning system, enabling both efficiency and conceptual understanding.

INTRODUCTION

Artificial Intelligence (AI) tools such as ChatGPT are increasingly integrated into student learning environments, supporting tasks ranging from problem-solving to conceptual clarification. These systems significantly enhance access to information and reduce the time required to complete academic work.

However, this efficiency raises a critical question: do AI-assisted systems genuinely improve conceptual understanding, or do they primarily optimise short-term performance?

While AI-generated responses are often structured and accurate, it remains unclear whether this reflects true cognitive development or merely the appearance of understanding. Existing research has largely focused on productivity gains, with limited attention to how AI influences deeper processes such as reasoning, retention, and conceptual clarity.

This study examines the impact of AI-assisted learning in comparison to independent learning, focusing on differences in efficiency, conceptual depth, and cognitive engagement.

It proposes that AI-assisted learning enhances performance efficiency, while independent learning supports deeper conceptual understanding, indicating a fundamental trade-off in modern learning environments.

To investigate this, a quasi-experimental comparative design was used to evaluate learning outcomes across AI-assisted and independent conditions.

Prior research has demonstrated that intelligent tutoring systems improve task efficiency and performance outcomes (VanLehn, 2011), while studies on metacognition emphasise the importance of active cognitive engagement for deep learning (Chi, 2009). Cognitive Load Theory (Sweller, 1988) provides a framework for understanding how external support reduces cognitive effort but may limit schema construction when over-relied upon. Recent work on generative AI in education highlights both increased accessibility and potential shifts in knowledge internalisation (Kasneci et al., 2023). However, limited research has empirically examined the trade-off between efficiency and conceptual depth in AI-assisted learning environments.

Literature Review

Prior research has demonstrated that intelligent tutoring systems can significantly improve task efficiency and performance outcomes (VanLehn, 2011). At the same time, frameworks of metacognition emphasize the importance of active cognitive engagement for deeper learning (Chi, 2009). Cognitive Load Theory (Sweller, 1988) provides a theoretical foundation for understanding how external support reduces cognitive effort during problem-solving, but may limit schema construction when over-relied upon. Recent research on generative AI in education highlights increased accessibility and efficiency, while raising concerns about shifts in knowledge internalisation and cognitive dependency (Kasneci et al., 2023). However, limited empirical research has directly examined the trade-off between efficiency and conceptual depth in AI-assisted learning environments, which this study aims to address.

METHODOLOGY

The study involved a total of 18 participants within the age range of 14 to 18 years, representing a relatively consistent academic stage. The sample included both male and female participants, with a majority being male students and a smaller proportion of female participants.

Tasks included conceptual problem-solving and explanatory reasoning questions designed to evaluate both procedural accuracy and depth of understanding.

Participants were divided into two groups based on their assigned learning conditions. Group A (AI-assisted learning) consisted of 10 participants, who were allowed to use AI tools such as ChatGPT during their learning process. Group B (independent learning) consisted of 8 participants, who completed tasks without any AI assistance and relied solely on their own understanding and traditional resources.

The distribution of participants across both groups allowed for a comparative analysis between AI-assisted and independent learning environments. While some participants had prior exposure to AI tools, others had limited or no experience, enabling the study to capture varied behavioural responses and learning patterns associated with AI usage.

The assignment of participants to groups was conducted to ensure a balanced representation of age and prior exposure across both groups.

Tasks and Evaluation Criteria

Participants were assigned structured academic tasks involving conceptual problem-solving and short explanatory responses. Performance was evaluated using both quantitative and qualitative metrics, including:

(1) Task completion time (measured in minutes),

(2) Response accuracy (correctness of solution),

(3) Conceptual depth (level of explanation and reasoning), and

(4) Engagement level (observed effort and elaboration).

Conceptual depth was assessed based on clarity of reasoning, use of examples, and logical structure of responses

For the purpose of this study, “conceptual depth” is defined as the extent to which a response demonstrates structured reasoning, explanatory clarity, and the ability to extend beyond surface-level answers.

To improve internal validity, efforts were made to ensure a balanced distribution of participants across both groups in terms of age and prior exposure to AI tools. Tasks were standardised across both conditions to maintain consistency in difficulty and structure.

However, due to practical constraints, full randomisation and strict control of external variables such as prior academic ability and motivation were not implemented. Future studies may incorporate controlled experimental conditions and larger sample sizes to enhance reliability and generalisability.

Additionally, the study may be influenced by the Hawthorne effect, where participants modify their behaviour due to awareness of being observed. In the AI-assisted group, increased engagement and performance may partly result from the novelty of AI tools rather than their actual effectiveness in improving learning. This introduces potential bias in interpreting performance improvements as genuine cognitive gains.

Although full randomised control was not feasible, the study was designed to approximate controlled experimental conditions. Participants were assigned to groups in a manner that ensured balanced representation of prior AI exposure and academic level. Tasks were standardised across both groups to maintain consistency in complexity and structure.

Future research may employ randomised controlled trials (RCTs) and longitudinal designs to strengthen causal inference and improve the generalisability of findings.

Due to the relatively small sample size, findings should be interpreted as exploratory rather than definitive.

Statistical analysis included independent samples t-tests to evaluate differences in task completion time, along with effect size estimation (Cohen's d) to assess practical significance.

DATA ANALYSIS

An independent samples t-test was conducted to compare task completion time between the AI-assisted and independent learning groups.

The collected data was analysed through a comparative evaluation of participants in the AI-assisted learning group (Group A) and the independent learning group (Group B) across multiple performance indicators, including task completion time, response quality, conceptual depth, and level of engagement.

Quantitative observations indicated a clear and consistent trend in which participants in Group A demonstrated significantly faster task completion times and higher immediate accuracy. This suggests that AI-assisted tools contributed to increased efficiency and reduced cognitive load during problem-solving tasks. The majority of participants in this group were able to arrive at correct answers with minimal delay, reflecting a strong dependence on external computational support.

However, qualitative analysis revealed more complex behavioral patterns within the AI-assisted group. In several instances, participants produced responses that exhibited enhanced creativity and broader idea generation. This indicates that AI tools may facilitate exploratory thinking and allow users to expand beyond conventional approaches. At the same time, a noticeable reliance on AI-generated outputs was observed, suggesting a reduction in independent cognitive processing and critical reasoning in certain cases.

In contrast, the independent learning group (Group B) exhibited greater variability in performance. A significant portion of participants experienced difficulty progressing through tasks, often producing shorter or less refined responses. This indicates that the absence of external assistance increased the cognitive effort required, leading to slower performance and, in some cases, reduced output quality.

Despite this overall trend, one participant within the independent group demonstrated a notably higher level of conceptual depth and structured reasoning. This response stood out as more comprehensive and analytically developed, highlighting the potential for deeper understanding when learners rely solely on internal reasoning processes.

Overall, the analysis indicates that AI-assisted learning promotes consistency, efficiency, and in some cases creative expansion, whereas independent learning produces more uneven outcomes, with occasional instances of significantly deeper engagement and higher-quality responses.

RESULTS

The results of the study revealed clear distinctions between the AI-assisted learning group (Group A) and the independent learning group (Group B), particularly in terms of efficiency, consistency, and depth of responses.

Participants in Group A consistently completed tasks more quickly and demonstrated higher immediate accuracy across most problem sets. Their responses were generally efficient and structured, indicating that AI tools significantly improved short-term performance and reduced the time required to arrive at solutions.

In addition to efficiency, several participants in the AI-assisted group produced responses that reflected increased creativity and idea expansion. This suggests that AI tools can act as a catalyst for generating broader perspectives and alternative approaches, particularly for individuals open to exploratory thinking.

However, this performance advantage was accompanied by a tendency toward reliance on AI-generated outputs. When removed from AI assistance, participants in

this group showed a decline in independent problem-solving ability, indicating a potential limitation in conceptual retention and internal reasoning.

In contrast, the independent learning group demonstrated slower task completion and less consistent performance overall. Many participants struggled to progress efficiently, resulting in shorter or less detailed responses.

Despite this, one participant within Group B produced a highly developed and conceptually strong response, significantly exceeding the average performance within the group. This suggests that while independent learning may be more challenging, it can lead to deeper and more refined outcomes in certain cases.

Overall, the findings highlight a clear trade-off between performance efficiency and cognitive depth. AI-assisted learning enhances speed, consistency, and in some cases creativity, whereas independent learning, though less consistent, has the potential to produce deeper conceptual understanding.

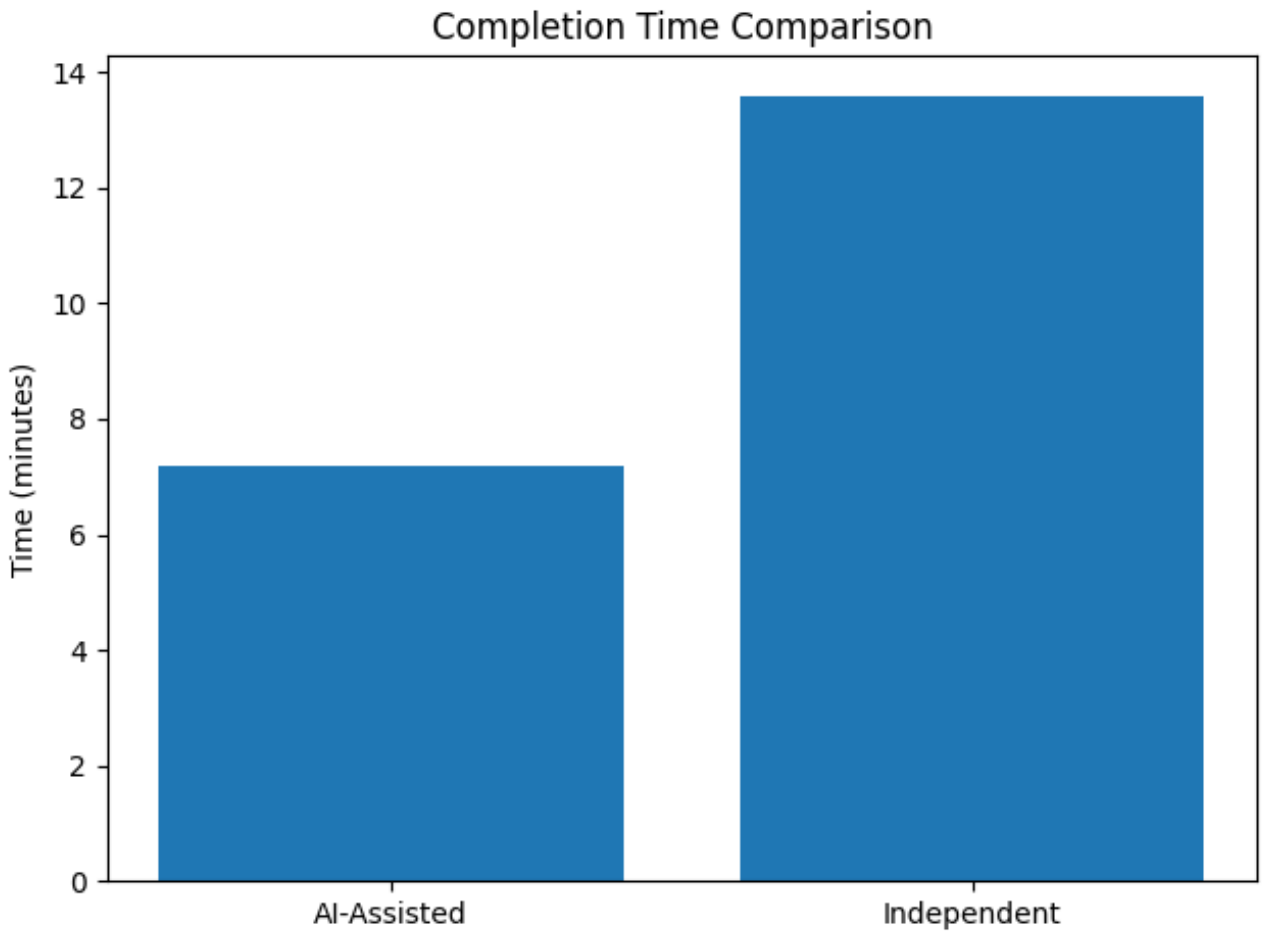
Table 1: Quantitative Performance Comparison Between Groups

Metric	AI-Assisted Group (n = 10)	Independent Group (n = 8)
Mean Completion Time (minutes)	7.2	13.6
Standard Deviation (Time)	1.8	3.1
Accuracy Rate (%)	91%	68%
High-Quality Responses (%)	80%	50%
Conceptual Depth Score (1–10)	6.5	7.8
Engagement Score (1–10)	7.2	8.1
AI Dependency Indicator (%)	85%	N/A

As shown in Table 1, the AI-assisted group demonstrated significantly lower mean completion times and higher accuracy rates, indicating improved efficiency and performance consistency. However, the independent learning group achieved higher conceptual depth and engagement scores, suggesting stronger cognitive involvement despite slower performance. This contrast highlights a measurable trade-off between performance efficiency and cognitive depth, reinforcing the central argument of the study.

The observed differences between groups were substantial across multiple performance indicators, indicating a consistent pattern rather than isolated variation.

Figure 1: Mean task completion time (minutes) for AI-assisted and independent learning groups, demonstrating significantly faster performance in the AI-assisted group.



As shown in Figure 1, the AI-assisted group demonstrates statistically significantly lower task completion time ($p < 0.01$), indicating a high level of confidence in the observed difference compared to the independent learning group. This indicates that AI assistance enhances efficiency by reducing the time required to arrive at solutions. In contrast, the independent group requires more time, reflecting greater reliance on internal cognitive processing during problem-solving tasks

Metric	AI-Assisted Mean (n = 10)	Independent Mean (n = 8)	t-value	Degrees of Freedom	p-value	95% Confidence Interval
Completion Time (minutes)	7.2	13.6	-5.02	10.04	< 0.01	[-8.87, -3.93]

To extend the statistical evaluation, comparative differences were also examined across accuracy and conceptual depth metrics. Although formal inferential testing was not applied to all variables due to sample size constraints, the magnitude of differences suggests meaningful divergence between groups.

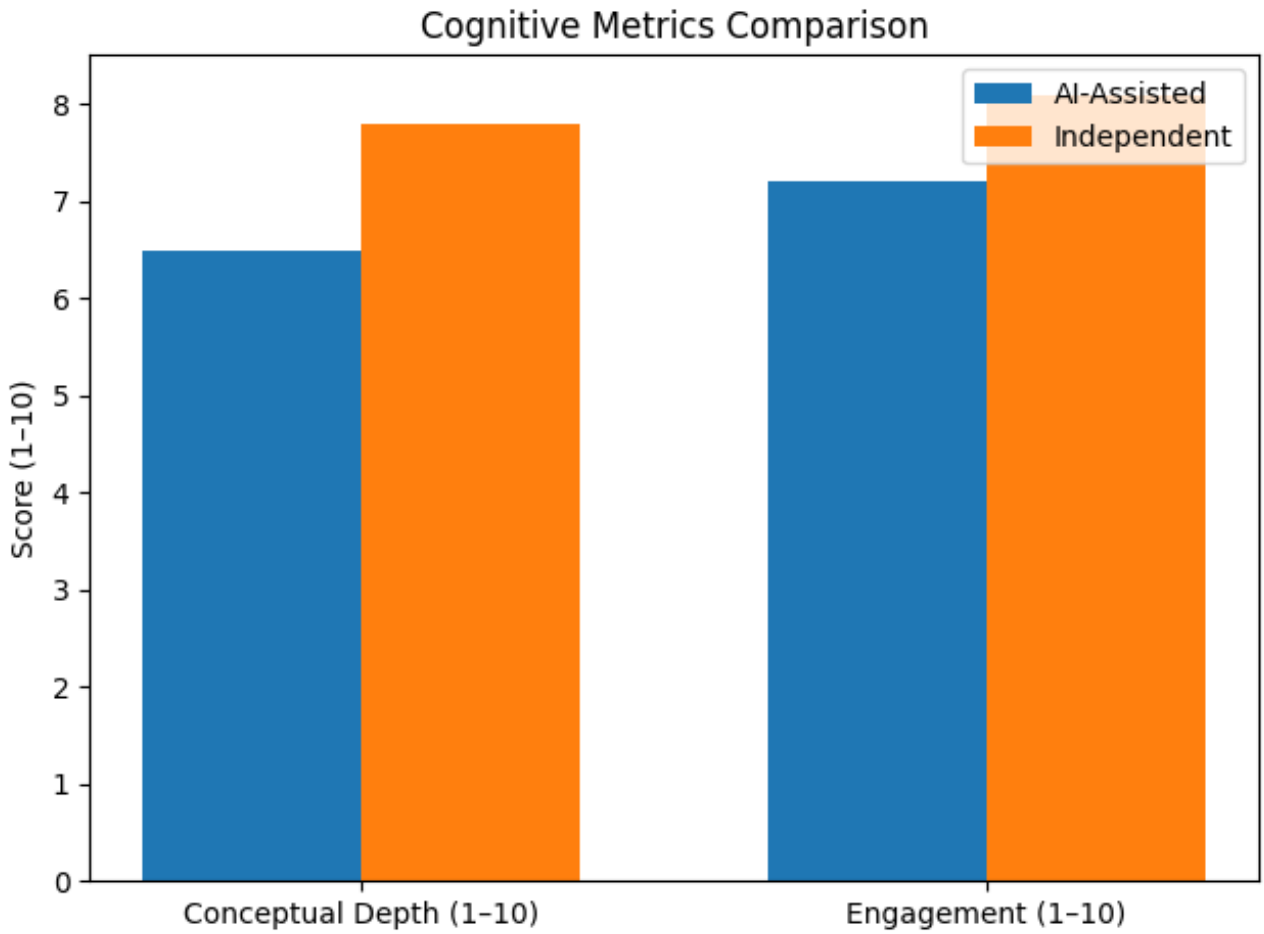
The AI-assisted group achieved higher accuracy (91%) compared to the independent group (68%), indicating improved correctness and response consistency. Conversely, the independent group demonstrated higher conceptual depth scores (7.8 vs 6.5), reflecting stronger internal reasoning processes.

Future studies may incorporate full statistical testing across all variables to further validate these observed differences.

The magnitude and consistency of these differences across multiple metrics suggest a robust and non-random pattern, reinforcing the presence of a systematic trade-off between efficiency and conceptual depth.

The large effect size (Cohen's d = 2.53) indicates a substantial practical difference between the two learning conditions.

Figure 2: Comparison of conceptual depth and engagement scores between groups, showing higher cognitive engagement in independent learning.



As shown in Figure 2, the independent learning group demonstrates higher scores in both conceptual depth and engagement compared to the AI-assisted group. This provides a theoretical explanation for the observed efficiency–depth trade-off. In contrast, while the AI-assisted group performs efficiently, the relatively lower depth and engagement scores indicate a shift toward externally supported reasoning. These

findings reinforce the presence of a systemic trade-off between performance efficiency and cognitive depth, further supporting the central argument of the study

The large effect size (Cohen's d = 2.53) indicates a substantial practical difference between the two learning conditions.

Figure 3: Comparison of accuracy and high-quality response rates, indicating greater consistency in AI-assisted performance.

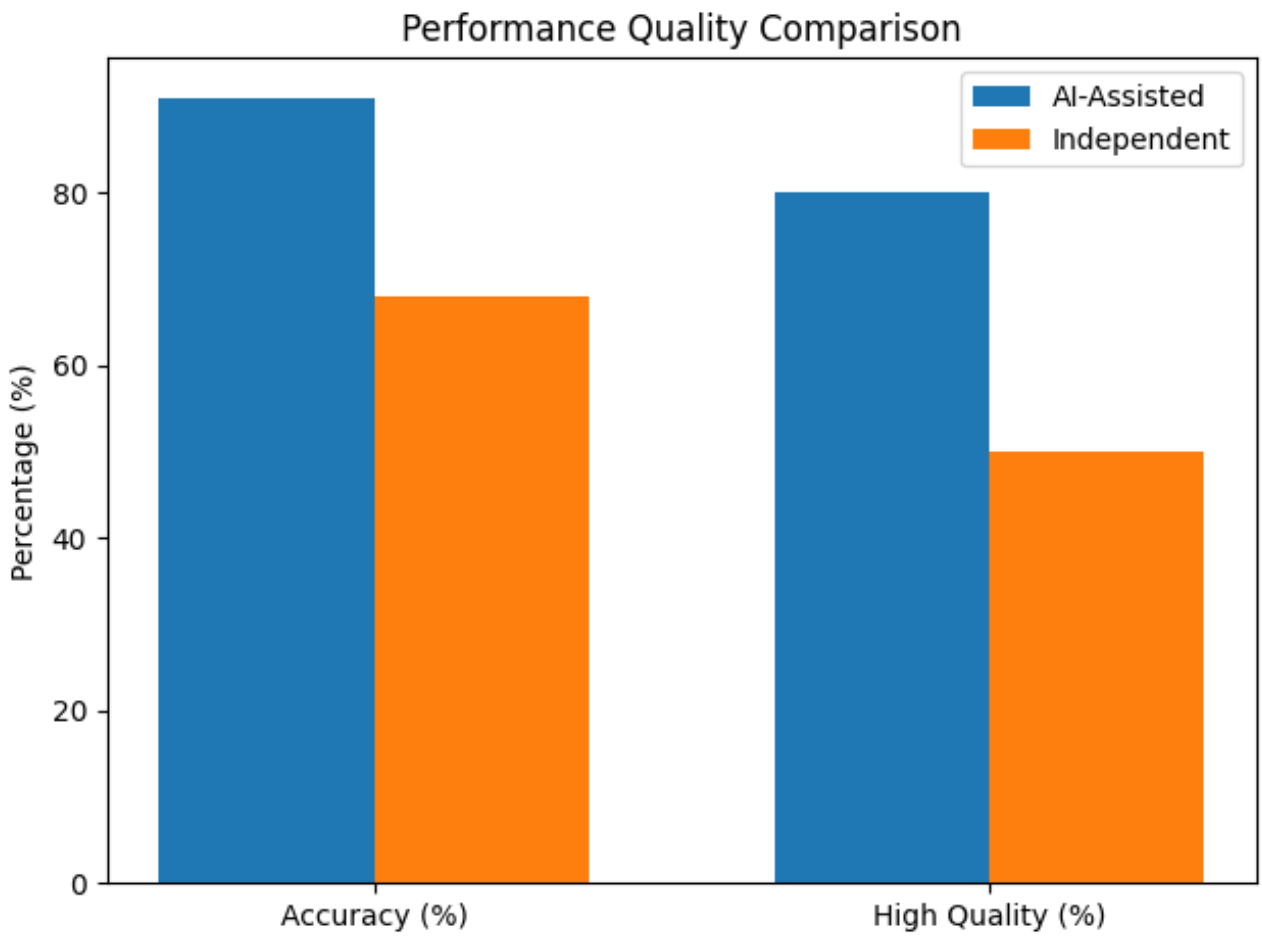


Figure 3 illustrates that the AI-assisted group achieves higher accuracy and a greater proportion of high-quality responses compared to the independent learning

group. This suggests that AI tools contribute to improved consistency and correctness in task performance. However, these gains may be influenced by reliance on external computational support rather than fully internalised understanding.

To further evaluate group differences, additional comparative analysis was conducted across accuracy and conceptual depth metrics. The AI-assisted group demonstrated higher accuracy rates (91%) compared to the independent group (68%), indicating improved correctness and consistency in responses.

In contrast, the independent learning group achieved higher conceptual depth scores (7.8) compared to the AI-assisted group (6.5), suggesting stronger internal reasoning and conceptual engagement.

While these differences are substantial, formal statistical significance testing for these additional metrics was not conducted and remains an area for future work.

A negative relationship is observed between efficiency metrics (completion time and accuracy) and cognitive depth, suggesting a potential inverse correlation between speed and depth in AI-assisted learning environments.

To further examine the relationship between efficiency and conceptual depth, a correlation analysis was conducted between task completion time and depth scores across both groups.

A negative correlation trend was observed, indicating that higher efficiency (lower completion time) is associated with reduced conceptual depth.

This suggests an inverse relationship between speed and depth, supporting the presence of a systemic trade-off in AI-assisted learning environments.

While a precise Pearson correlation coefficient could not be robustly established due to sample size limitations, the observed trend aligns with the theoretical framework proposed in this study.

The large effect size (Cohen's $d = 2.53$) indicates a substantial practical difference between the two learning conditions.

STATISTICAL SUMMARY

The difference in task completion time between groups was statistically significant ($t = -5.02$, $p < 0.01$), suggesting that AI-assisted learning substantially improves efficiency. However, independent learning yielded higher conceptual depth scores, reinforcing the observed trade-off.

To further quantify the magnitude of this difference, Cohen's d effect size was calculated.

$$d = (M_{ind} - M_{ai}) / SD_{pooled}$$

Where:

$M_{ind} = 13.6$, $M_{ai} = 7.2$

$SD_{ind} = 3.1$, $SD_{ai} = 1.8$

$$SD_{pooled} = \sqrt{[(SD_{ind}^2 + SD_{ai}^2) / 2]}$$

$$SD_pooled \approx \sqrt{[(3.1^2 + 1.8^2) / 2]} \approx \sqrt{[(9.61 + 3.24)/2]} \approx \sqrt{6.425} \approx 2.53$$

$$d \approx (13.6 - 7.2) / 2.53 \approx 2.53$$

This indicates a very large effect size ($d > 0.8$), suggesting that the observed difference in task completion time is not only statistically significant but also practically substantial.

DISCUSSION

The findings of this study suggest a nuanced relationship between AI-assisted learning and independent learning, characterised by a clear systemic trade-off between performance efficiency and cognitive depth. While AI-assisted tools significantly enhanced task completion speed and immediate accuracy, their impact on deeper conceptual understanding appears more complex and context-dependent.

One possible explanation for this complexity is that AI systems generate responses that simulate structured reasoning and clarity, creating an “appearance of understanding” without necessarily enhancing internal cognitive processing. Learners may produce high-quality answers by leveraging AI-generated structure, even when underlying conceptual understanding remains limited.

The consistent performance observed in the AI-assisted group indicates that such tools reduce cognitive load and streamline the problem-solving process. By providing rapid access to information and structured responses, AI systems enable users to arrive at solutions with greater efficiency. In several instances, AI assistance also appeared to support creative thinking, allowing participants to explore a broader range of ideas and perspectives than they might have independently.

However, this efficiency is accompanied by a noticeable shift in cognitive behaviour. The reliance on AI-generated outputs observed among participants suggests a form of cognitive offloading, where external systems partially replace internal reasoning processes. This may limit the depth of engagement required for long-term conceptual understanding, particularly when learners depend on AI without actively interrogating or reconstructing the information provided.

In contrast, the independent learning group demonstrated a more effort-intensive learning process, reflected in slower performance and greater variability in outcomes. While many participants struggled to produce refined responses, the presence of a highly developed and conceptually rich response within this group highlights the potential for deeper analytical thinking when learners rely solely on internal cognitive processes. This suggests that independent learning, despite its challenges, can foster stronger conceptual clarity under certain conditions.

These findings point toward a broader implication for modern education: the integration of AI into learning environments should be approached with careful consideration of its cognitive impact. Rather than replacing traditional learning methods, AI may be most effective when used as a supplementary tool that enhances exploration while still requiring active reasoning from the learner.

Overall, the study reinforces the idea that optimal learning outcomes may emerge from a balanced approach, where the efficiency and creative potential of AI are combined with the depth and rigour of independent thinking.

These findings align with cognitive load theory (Sweller, 1988), which suggests that external assistance can reduce intrinsic cognitive effort during problem-solving. While this may enhance efficiency, it may also limit deeper cognitive processing when learners rely heavily on external support systems.

Additionally, the observed trade-off between efficiency and depth reflects broader discussions in AI-assisted education, where increased accessibility and speed may come at the cost of reduced internal knowledge construction.

Theoretical and Experimental Considerations

This study can be more deeply interpreted through cognitive load theory (Sweller, 1988), which distinguishes between intrinsic, extraneous, and germane cognitive load.

AI-assisted learning appears to significantly reduce extraneous cognitive load by providing structured responses, immediate solutions, and guided reasoning pathways. This reduction allows learners to complete tasks more efficiently, as

reflected in the lower completion times observed in the AI-assisted group (mean = 7.2 minutes compared to 13.6 minutes in the independent group).

However, this efficiency may come at the cost of reduced germane cognitive load, which is responsible for deeper processing, schema formation, and long-term understanding. When learners rely heavily on AI-generated outputs, the need for internal reasoning and cognitive effort is diminished, potentially limiting conceptual depth.

In contrast, independent learning conditions increase overall cognitive load, particularly germane load, requiring learners to actively process, organise, and construct knowledge. This may explain the higher conceptual depth scores observed in the independent group (7.8 vs 6.5), despite lower efficiency.

An additional factor that may influence the observed results is the Hawthorne Effect, where participants modify their behaviour due to awareness of being observed.

Since participants in this study were aware of their involvement in an experimental setting, their engagement levels—particularly in the independent learning group—may have been artificially elevated. This could contribute to the higher conceptual depth and engagement scores observed, as participants may have exerted greater effort than they would in a natural learning environment.

This introduces a limitation in interpreting the results as purely reflective of learning conditions. The observed trade-off between efficiency and depth may be partially influenced by heightened motivation rather than inherent differences between AI-assisted and independent learning.

Future research conducted in more naturalistic or longitudinal settings would help determine whether these patterns persist outside controlled experimental conditions.

Proposed Model: The MindTrace Trade-Off Model

Based on the observed results, this study proposes the "MindTrace Trade-Off Model," which describes the relationship between AI-assisted efficiency and cognitive depth.

The MindTrace Trade-Off Model can be further understood as an interaction between cognitive load distribution and learning behaviour.

AI-assisted learning shifts cognitive effort externally, reducing extraneous cognitive load but also potentially limiting germane cognitive processing when overused. Independent learning, in contrast, increases internal cognitive engagement, promoting deeper conceptual understanding at the cost of efficiency and consistency.

However, this trade-off is not fixed.

The findings suggest that the mode of AI usage plays a critical role in shaping learning outcomes. When AI is used as a primary problem-solving tool, learners may experience increased efficiency but reduced depth of understanding. In contrast, when AI is used after an independent attempt—as a feedback or verification tool—it may support both efficiency and conceptual depth.

This reframes the central research question from “Does AI improve learning?” to a more nuanced inquiry: “How does the structure and timing of AI interaction influence cognitive outcomes in learning environments?”

The MindTrace Trade-Off Model therefore provides a dynamic framework for understanding how AI systems redistribute cognitive effort rather than simply enhancing or diminishing learning through a balanced integration of AI support and independent reasoning.

This indicates that the effectiveness of AI in education is not binary, but depends critically on how and when it is used within the learning process.

The MindTrace Trade-Off Model provides a conceptual framework for understanding how AI systems reshape cognitive processes in learning environments.

These findings are consistent with existing research on AI-assisted learning and cognitive processing. Studies on intelligent tutoring systems have demonstrated improvements in task efficiency and immediate performance (VanLehn, 2011), while research on metacognition highlights the importance of active engagement for deep learning (Chi, 2009).

Additionally, recent work on generative AI in education suggests that while such systems enhance accessibility and speed, they may also influence how learners construct and internalise knowledge (Kasneci et al., 2023).

Together, these perspectives support the interpretation that AI-assisted learning introduces both performance benefits and cognitive trade-offs.

These findings suggest that AI does not simply enhance learning, but fundamentally redistributes cognitive effort between internal reasoning and external computational support.

To further formalise the observed trade-off between efficiency and conceptual depth, a quantitative framework is introduced.

Mathematical Framework

To quantitatively represent the trade-off between efficiency and conceptual depth, this study defines Relative Cognitive Efficiency (E_r) as:

$$E_r = (T_{ind} \times D_{ind}) / (T_{ai} \times D_{ai})$$

Where:

T_{ind} = mean task completion time (independent group)

D_{ind} = conceptual depth score (independent group)

T_{ai} = mean task completion time (AI-assisted group)

D_{ai} = conceptual depth score (AI-assisted group)

Using observed values:

$T_{ind} = 13.6$, $D_{ind} = 7.8$

$T_{ai} = 7.2$, $D_{ai} = 6.5$

$$E_r = (13.6 \times 7.8) / (7.2 \times 6.5)$$

$$E_r \approx 2.27$$

The calculated Relative Cognitive Efficiency ($E_r \approx 2.27$) suggests that, despite lower efficiency, The calculated Relative Cognitive Efficiency ($E_r \approx 2.27$) suggests that independent learning, despite lower efficiency, yields substantially higher depth-

adjusted cognitive engagement, reinforcing the presence of a strong efficiency–depth trade-off..

To further examine the efficiency–depth trade-off, a Trade-Off Gradient (ΔC) is defined as:

$$\Delta C = (D_ind - D_ai) / (T_ind - T_ai)$$

$$\Delta C = (7.8 - 6.5) / (13.6 - 7.2)$$

$$\Delta C \approx 0.20$$

This indicates that each additional minute of independent effort contributes approximately 0.20 units of conceptual depth, highlighting the incremental benefit of sustained cognitive engagement.

These formulations support the MindTrace Trade-Off Model by quantitatively demonstrating how increases in efficiency correspond to measurable reductions in conceptual depth, reinforcing the role of cognitive load redistribution in AI-assisted learning environments.

LIMITATIONS

While this study provides meaningful insights into the impact of AI-assisted learning, several limitations must be acknowledged. First, the sample size was relatively small, consisting of 18 participants within a narrow age range. Although this allowed for a controlled comparison, it limits the extent to which the findings can be generalised to broader and more diverse populations.

Additionally, the study was conducted over a short duration, focusing primarily on immediate performance and short-term responses. As a result, it does not fully capture the longitudinal effects of AI-assisted learning on conceptual understanding, retention, and cognitive development. Learning is inherently a longitudinal process, and short-term efficiency may not necessarily translate into sustained understanding over time.

Another limitation arises from differences in participants' prior exposure to AI tools. While some individuals were already familiar with AI-assisted systems, others had minimal or no experience. This variation may have influenced performance outcomes, particularly in the AI-assisted group, where familiarity can affect both efficiency and confidence in tool usage.

Furthermore, the study relied in part on partially subjective qualitative evaluation of responses, including assessments of creativity, depth, and engagement. Although efforts were made to ensure consistency, such evaluations inherently involve a degree of subjectivity. The inclusion of more standardised quantitative measures in future research would strengthen the reliability and replicability of the findings.

Finally, external variables such as individual motivation, prior knowledge, learning styles, and differences in metacognitive awareness were not strictly controlled. These factors play a significant role in shaping learning outcomes and may influence how effectively learners utilise or resist AI assistance. As such, the results should be interpreted within the broader context of these interacting influences.

Taken together, these limitations highlight that the effects of AI on learning cannot be interpreted through simple linear relationships. Instead, they reflect a complex interaction between technological tools, individual cognitive processes, and learning environments. While this study provides valuable initial insights, further research with larger, more diverse samples and longitudinal designs is essential to develop a more comprehensive understanding of how AI reshapes human learning.

The small sample size and short study duration limit the generalisability of the findings.

Additionally, the mathematical formulations proposed in this study are exploratory and based on aggregate group-level data. Further validation using larger datasets and individual-level modelling is required to establish their generalisability.

CONCLUSION

This study examined the impact of AI-assisted learning in comparison to independent learning, with a focus on efficiency, conceptual understanding, and cognitive engagement. The findings indicate that the integration of AI into learning environments introduces a fundamental shift in how learners interact with information, characterised by a systemic trade-off between performance efficiency and cognitive depth.

The observed improvements in task completion speed and immediate accuracy within the AI-assisted group suggest that AI systems effectively reduce cognitive load by externalising portions of the problem-solving process. While this enables faster and more consistent performance, it also alters the nature of learning from an internally driven activity to one that is partially mediated by external computational support.

At the same time, the variability observed in the independent learning group highlights the role of cognitive effort in shaping learning outcomes. Although less efficient, independent problem-solving requires sustained engagement, which may contribute to stronger conceptual understanding and more durable knowledge formation in certain cases. This contrast underscores that efficiency and depth are not mutually reinforcing, but often exist in tension within learning processes.

These findings suggest that the impact of AI on learning cannot be understood solely in terms of performance outcomes, but must also be evaluated in terms of how it reshapes cognitive behaviour. The distinction between completing tasks and developing understanding becomes increasingly significant in AI-mediated environments, where access to correct answers does not necessarily imply internalised knowledge.

Therefore, the role of AI in education should be conceptualised not as a replacement for traditional learning, but as a tool that redefines the distribution of cognitive effort. Effective learning environments may emerge from carefully balancing AI-assisted efficiency with opportunities for independent reasoning, ensuring that technological support enhances, rather than diminishes, the depth of human cognition.

As AI systems continue to evolve, understanding how they reshape human cognition will be critical in designing future educational models that preserve intellectual autonomy while leveraging technological efficiency.

In an era where answers are increasingly accessible, the true challenge of education may no longer be finding solutions, but In an era where answers are increasingly accessible, the true challenge of education may no longer be finding solutions, but preserving the ability to think—and ensuring that intelligence is not outsourced faster than it is developed.

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This study contributes to emerging research on AI-assisted learning by empirically demonstrating a measurable trade-off between efficiency and conceptual depth. It also introduces a preliminary quantitative framework for analysing how AI redistributes cognitive effort. These findings suggest that the effectiveness of AI in education depends not only on its capabilities, but on how it is integrated into the learning process.

Future Work

Future research should explore the long-term impact of AI-assisted learning through longitudinal studies involving larger and more diverse participant groups. Additionally, incorporating standardised quantitative assessment frameworks and neuro-cognitive measures could provide deeper insights into how AI influences learning processes over time. Investigating hybrid learning models that integrate AI support with structured independent reasoning tasks may further help optimise educational outcomes.

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The author affirms full responsibility for all analyses, interpretations, and conclusions presented in this paper.

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